

Methods

fNIRS Data Acquisition

fNIRS data were recorded using a NIRXport2 8-8 Core Unit (Brain Products GmbH) operating at a native sampling rate of 4.36 Hz. The device used two wavelengths (760 nm and 850 nm) and a total of 25 source-detector channel pairs distributed across the left hemisphere. The optode array covered two regions of interest (ROIs): the left frontal cortex (7 channels: S1-D1, S2-D1, S2-D3, S1-D2, S3-D3, S4-D1, S3-D1) and the left temporal cortex (9 channels: S8-D7, S8-D4, S7-D7, S7-D4, S7-D6, S6-D4, S5-D5, S5-D6, S5-D4; S6-D5 was excluded due to insufficient signal quality). Data were recorded in a naturalistic classroom setting across all ten sessions (one calibration session and nine training sessions) of the ACELETRA(R) programme.

Stimulus Event Reconstruction

The ACELETRA(R) software generated multiple raw stimulus channels encoding individual trial-level events that were not directly suitable for block-level general linear model (GLM) analysis. A custom MATLAB script (`contagem_nova.m`) was therefore developed to reconstruct block-level stimulus events from the raw onset markers prior to first-level GLM estimation. For each participant-session pair, reading activity channels (carrying stimulus onset and offset markers) were identified and distinguished from rest channels (inter-block periods), the latter being detected automatically as the channel containing the fewest recorded events.

Block boundaries were defined by the onset times of rest periods, such that each block was delimited by two consecutive rest markers. Within each block, the epoch onset was set to the timestamp of the first reading stimulus, and the epoch duration was computed as the difference between the timestamp of the last reading offset and that onset. Amplitude was set to unity (boxcar regressor). The expected block structure comprised four blocks per calibration session and seven blocks per training session (`sessao_num = 1-9`), each containing five reading stimuli. Automated integrity checks compared the reconstructed block count and stimulus count against expected values; deviations triggered runtime warnings and were flagged for manual inspection. Following reconstruction, all original stimulus channels were discarded (`nirs.modules.DiscardStims`) and a single compound stimulus object labelled "Aceletra" (`nirs.design.StimulusEvents`) was assigned to each data object, replacing the original multi-channel structure.

fNIRS Data Preprocessing

Preprocessing was performed in MATLAB using the NIRS Toolbox (`pre_processamento.m`). The pipeline comprised seven sequential steps applied to all 140 session objects (14 participants x 10 sessions):

- (1) Resampling: raw signals were resampled to a uniform rate of 4 Hz (`nirs.modules.Resample`, `Fs = 4`).
- (2) Baseline trimming: a 10-second post-stimulus baseline was retained following each trial onset, with zero pre-stimulus padding (`nirs.modules.TrimBaseline`, `preBaseline = 0` s, `postBaseline = 10` s).
- (3) Optical density conversion: raw light intensity signals were converted to changes in optical density (`nirs.modules.OpticalDensity`).
- (4) Modified Beer-Lambert law: chromophore concentration changes were estimated via the modified Beer-Lambert law (`nirs.modules.BeerLambertLaw`), yielding time series for oxyhaemoglobin (HbO), deoxyhaemoglobin (HbR), and total haemoglobin (HbT).
- (5) Total haemoglobin calculation: HbT was computed as the sum of HbO and HbR (`nirs.modules.CalculateTotalHb`).
- (6) Band-pass filtering: a zero-phase Butterworth band-pass filter was applied to retain haemodynamic frequencies between 0.01 and 0.10 Hz (`nirs.modules.BandPassFilter`, `highpass = 0.01` Hz, `lowpass = 0.10` Hz, `do_downsample = 0`), without additional downsampling after filtering.
- (7) Wavelet-based motion artefact correction: motion artefacts were attenuated using a wavelet-based filter (`nirs.modules.WaveletFilter`). Because this module does not propagate stimulus information, stimulus objects were manually re-assigned from the pre-wavelet data objects to the filtered outputs after this step.

First-Level General Linear Model

Following preprocessing, a first-level GLM was estimated for each session using `nirs.modules.GLM` with the short-separation channel regressor option enabled (`AddShortSepRegressors = 1`). Short-separation channels record primarily scalp-level physiology; including them as nuisance regressors reduces the influence of superficial signal contamination on the haemodynamic response estimates. The single task regressor was the reconstructed block-level "Aceletra" stimulus object. The resulting subject-level GLM objects (`SubjStats`) contained one beta coefficient per channel per chromophore per session, yielding 140 `SubjStats` objects in total (14 participants x 10 sessions).

Pipeline Quality Control

Pipeline integrity was verified using a dedicated MATLAB quality-check script (`relatorio_geral.m`) that executed automated checks across five domains. (1) Basic structure: total session count (expected: 140), sampling rate (expected: 4 Hz across all sessions), four chromophores present (StO₂, HbO, HbR, HbT), and channel count (expected: 100 = 25 channels x 4 chromophores). (2) Data quality: zero NaN and Inf values across all sessions; HbO amplitude within the expected range ([p1, p99] within +/- 500 uM); absence of flat-signal sessions (standard deviation < 0.001). (3) Stimulus integrity: stimulus object present in all sessions; stimulus name = "Aceletra"; block

counts consistent with expected values (4 or 7 per session); block durations within 5-600 s. (4) Demographic consistency: group labels valid (acelerado or nao_acelerado); 10 sessions per subject; the 14 expected participant identifiers present. (5) GLM consistency: 140 SubjStats objects present; condition "Aceletra" present in all; zero NaN betas. A summary report listed all critical errors and warnings.